

A novel Sequence to Sequence Data Modelling based CNN-LSTM Algorithm for Three Years Ahead Monthly Peak Load Forecasting

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Abstract— Long-term load forecasting (LTLF) models play an important role in the strategic planning of power systems around the globe. Obtaining correct decisions on power network expansions or restrictions based on predictions help substantially reduce the power grid infrastructure costs. The classical approach of LTLF is limited to the usage of artificial neural networks (ANN) or regression-based approaches along with a large set of historical electricity load, weather, economy and population data. Considering the drawbacks of classical methods, this paper introduces a novel sequence to sequence hybrid convolutional neural network and long short-term memory (CNN-LSTM) model to forecast the monthly peak load for a time horizon of three years. These drawbacks include, lack of sensitivity to changing trends over long time horizons, difficulty of fitting large number of variables and complex relationships, etc. [1]. Forecasting time interval plays a key role in LTLF. Therefore, using monthly peak load avoids unnecessary complications while providing all essential information for a good long-term strategical planning. The accuracy of the proposed method is verified by the load data of “New South Wales (NSW)”, Australia. The numerical results show that, proposed method has achieved higher prediction accuracy compared to the existing work on long-term load forecasting.

Index Terms—Monthly Peak Load, Customer Load/Demand, Long-term Load Forecasting, Point Forecasting, Probabilistic Forecasting, Sequence to Sequence Modelling, Convolutional Neural Networks, Long Short Term Memory.

I. INTRODUCTION

The main aim of power companies is to provide their customers with a sufficient and reliable power supply. Therefore, power system operation and planning rely heavily on the load forecasting. Power network operators such as Western Power (WP), Ausgrid, SA (South Australia) Power Networks, etc. invest massively on forecasting studies to ensure that they have an understanding of the future of their power networks. Typically, based on the changes happening in customer demand patterns the forecasting work can be divided into three main categories, including short-term load forecasting (STLF), medium-term load forecasting (MTLF) and long-term load forecasting (LTLF). If the considered forecasting time horizon is intra-day or day-ahead, then it is considered to be STLF. Forecasting for the next few days, weeks or months is known as MTLF. Finally, LTLF covers forecasting horizons of one or more years ahead. Both STLF and MTLF solutions are used to make decisions in day to day power system operations, while LTLF aims to assist with power system infrastructure planning and budget allocations. Current literature reports extensive work on STLF and MTLF, however, compared to that, work reported on LTLF is relatively sparse.

According to the industrial practice, utility companies generally categorise the long-term planning work as; general tactical planning (few weeks to few months ahead), long-term tactical planning (1-3 years ahead) and strategic long-term planning (10-20 years ahead). They usually assign the long-term planning work in different time domains to separate research groups [2]. In this work, attention is paid towards this second category considering the importance of it towards efficient power system operation. Operators may need to plan the network operation if it is in a transitional stage during construction of new assets, similar to what is currently happening due to the overwhelming capacity increase of rooftop photovoltaic (PV). A simple non-power system analogy can be a free-way widening project which often results in long duration of congestion. Similarly when we reconfigure and/or build new transmission lines to support the connection of new facilities (generation or load), there will be long duration of periods which the project will put the network in different transitional stages, often the duration of these transitional stages can last weeks to a few months. Having a three year forecast of customer demand will allow analysis of the congestion caused by these transitional stages and allow development of mitigation strategies to resolve those identified issues. If we don't plan well during these congestion periods, then the users (i.e., owners of existing facilities) connected to the system may be impacted financially.

Apart from the afore-mentioned auxiliary components such as battery energy storage system (BESS) and backup generator planning, maintenance scheduling also benefits hugely by using monthly peak load data for a period of three years. Specially, when maintenance scheduling costs are considered, details about customer usage patterns, power, and energy fluctuations, etc. are needed, where yearly peak load values do not provide much insight on, and hourly, daily or weekly forecasting is too frequent considering the computational intensity of that in the long run [3]. On the other hand, monthly peak load values provide much more details about these usage patterns as well as not consume a lot of computational resources and time compared to the hourly, daily or weekly models, which will eventually result in overall cost reductions [4]. In summary, it can be said that hourly, daily or weekly load forecasting adds significant benefits to short/medium-term forecasting, but not in long-term forecasting.

LTLF work can be broadly categorised into two streams based on the calculation procedure: point forecasting methods and probabilistic forecasting methods. Considering the goal of this work, the literature review first introduces the existing point forecasting methods and then, discusses about the probabilistic forecasting methods. Compared to the STLF, LTLF

work has been spread across a great variety of time horizons and time intervals subject to the planning requirements of different network operators [5]. Dong and Grumbach [3] have categorised the LTLF methods into three groups as: top-down forecasting, bottom-up forecasting and hybrid forecasting. In here they have introduced a hybrid forecasting method based on LSTM/GRU sequence prediction techniques to forecast the yearly peak load for a limited forecasting horizon of one year and was able to achieve a Mean Absolute Percentage Error (MAPE) of 5.82%. Utility companies can reasonably plan long-term infrastructure upgrades and modifications only after proper forecast and assessment, therefore, having a forecasting time horizon of only one year limits its applicability to long-term network planning activities. Li et al. [6] have introduced a novel system dynamic (SD) and ImPACT equation based method to forecast long-term electricity consumption. Improved yearly consumption prediction results are demonstrated by dividing the driving factors of electricity consumption into four subsystems to reveal the connection between influencing factors and electricity consumption. However, obtaining the total power consumption throughout a year does not provide much insights to the long-term tactical planning of power systems (i.e., plan to operate the network if it is in a transitional stage during construction of new assets), rather predicting the monthly peak power consumption provides insights on under how much stress the network is currently undertaking.

Ali et al. [7] have introduced a fuzzy logic approach to forecast the monthly average load for a period of one year. Average load does not account for the extreme scenarios that may occur in a month, which is a major drawback of this work. Even by considering monthly average load the model was only able to achieve a MAPE of 6.9%, which still needs to be improved. From a planning perspective, it is very important to study the peak load of a month to take the critical decisions based on the worst case scenario that might take place. Agrawal et al. [8] implemented a long short term memory (LSTM) based hourly load prediction model for a planning horizon of five years. Nevertheless, it is less attractive, because obtaining long-term historic data with hourly interval is difficult as well as it does not add a significant value towards the long-term planning work. In fact, hourly load predictions are quite useful in STLF and MTLF, where we look at the frequent changes that happens in customer load pattern. However, using such an interval makes the training process more computationally intensive even though it helps improve the training accuracy. Noticeably, the obtained MAPE value in [8] was as high as 6.54%. A deep stacked LSTM model has been introduced in [9] to forecast the peak daily load of SA for a period of one year. Even though authors have recognised the importance of forecasting the daily load under LTLF, it is quite debatable whether daily peak load really matters to the long-term decisions for network planning. When it comes to the strategical planning of power systems, (i.e., designing of BESS or expansion of transmission/network connections) obtaining the monthly peak load values that account for the seasonal changes is accurate enough [10]. Daneshi et al. [10] introduced a fuzzy logic incorporated artificial neural network (ANN) hybrid model to

forecast the monthly peak load for a period of three years. It reported a MAPE value of 5.73% in the real world application in ISO New England market. It is understood that this result can be further improved in a deep neural network (DNN) model by considering different uncertainties that have a strong relationship with the load. After analysing the existing work, it can be concluded that, majority of existing LTLF work have shown key drawbacks in: selection of the most appropriate load data for the planning purpose, picking the correct time interval/horizon, etc. In our work these key deficiencies have been addressed along with model performance improvements.

In addition to the afore-mentioned point forecasting techniques, probabilistic forecasting methods have also been discussed in the literature for LTLF. Probability of exceedance (POE) is a key term that is commonly used in the probabilistic forecasting work, where it stands for the likelihood a maximum or minimum load forecast will be met or exceeded [11]. It can also be defined as the problem of estimating the probability that a time series will exceed a predefined threshold in a predefined future period. Li et al. [12] used extreme value theory to forecast the long-term substation peak electricity load. Designated POE levels have been used to generate the expression of probabilistic peak load. Authors state that it provides a way to forecast substation peak load that influences the stability and explanatory ability of energy, customer and PV forecasts. Hyndman et al. [13] have simulated the temperatures using a new double seasonal bootstrapping method to obtain a LTLF method that captures the complex non-linear effects of temperature. This has obtained good forecasting results for annual/weekly peak load up to ten years. Hong et al. [14] generated different weather scenarios and three macroeconomic scenarios to forecast the monthly energy and peak load for a period of one year. In summary, compared to the point forecasting methods, probabilistic forecasting methods have the ability to account for the uncertainties of variables in a long-term time horizon. Collectively, these work have demonstrated good results under 10%, 50% and 90% POE values, however, in majority of the published work, scenario generation has been limited to a single factor such as weather or economy. Identifying this drawback, in the second half of the paper, authors have introduced a novel block bootstrapping technique to generate multiple scenarios related to weather, economic and social factors.

Selecting the forecasting interval and the time horizon correctly is a fundamental component of any forecasting model. In theory, a good trade-off between the forecasting interval and time horizon contributes to enhance the long-term forecasting accuracy while ensuring its usefulness for network planning. This paper proposes an enhanced hybrid convolutional neural network (CNN) and LSTM (CNN-LSTM) based framework for LTLF, considering the excellent outcomes achieved by CNN-LSTM in the field of load forecasting. This work was inspired by the encoder-decoder LSTM network model introduced by Sutskever et al. [15]. Encoder-decoder LSTM network model, which is a sequence to sequence (seq2seq) model, was initiated for the purpose of mapping fixed-length inputs towards fixed-length outputs. Li et al. [16] demonstrated good MTLF prediction results by implementing a seq2seq

based LSTM network model for hourly load forecasting for a period of one month. In [17] a LSTM based DNN forecasting framework with appliance consumption sequences is proposed to address the volatility in residential load forecasting. Kong et al. implemented a sequential based LSTM framework for the purpose of short-term residential load forecasting. Compared to the recurrent neural networks (RNN), seq2seq models are much more stable and solves the problems like vanishing and exploding gradient. They also combine the advantages of univariate trend and multivariate regression models [3]. Also, seq2seq models support inputs and outputs with multiple features, whereas techniques like autoregressive integrated moving average (ARIMA) does not support this. In addition, this work was also inspired by the work done on financial time-series forecasting by combining different algorithms together to achieve a better performance [18]. Alhussein et al. [19] implemented a hybrid CNN-LSTM model for the purpose of short-term individual household load forecasting and validated it using the Smart Grid Smart City (SGSC) project data funded by the Australian Government. Inspired by these work, a CNN-LSTM architecture based encoder-decoder, seq2seq model was implemented in this work to study the monthly peak load forecasting for a period of three years. To avoid any complexity, hereafter, the hybrid CNN-LSTM model is introduced as just CNN-LSTM. The model's applicability and accuracy are confirmed by using customer load data from New South Wales (NSW), Australia.

In summary, application of CNN-LSTM framework can significantly improve the prediction accuracy compared to the existing work which have used statistical or neural artificial intelligence techniques. Combining CNN-LSTM avoids the drawbacks of each algorithm and introduce novelty into the forecasting work. The main contributions of this research can be summarized as follows:

- 1) Authors have developed a novel real world transmission system long-term tactical planning model to significantly increase the accuracy of monthly peak load forecasting, where it is shown that utilising PV power generation pattern and medium-term effects of continuously high temperature as important inputs can help improve the model accuracy.
- 2) A model based on CNN-LSTM is proposed for the purpose of long-term point load forecasting up to three years ahead. Application of this model to NSW load data along with the most important input features (features that have a direct relationship with monthly peak load) resulted in highly accurate long-term forecasting results. As per the best knowledge of authors, this work is the first of its kind in application of CNN-LSTM neural network technique in long-term electricity load forecasting.
- 3) Probabilistic load forecasting plays a key role in long-term tactical and strategical planning. Therefore, a block bootstrapping (i.e., for scenario generation based on economic, population and climate data) based long-term probabilistic load forecasting framework is developed in this work to highlight the importance of scenario generation to achieve highly accurate probabilistic long-

term tactical planning results.

The rest of paper is structured as follows: Section II presents the mathematical foundation of different neural networks. Section III introduces the proposed CNN-LSTM model. Section IV explains the process of CNN-LSTM model training, validation and testing. Section V discusses the application of the model in point forecasting. Section VI discusses the application of the model in probabilistic forecasting. Section VII and VIII concludes the paper and discusses about the future work.

II. INTRODUCTION TO NEURAL NETWORK MODELS

The aim of this section is to establish the mathematical foundation of the proposed method, where RNN, LSTM, CNN and CNN-LSTM architectures have been introduced.

A. Recurrent Neural Networks

RNN can be introduced as a collection of feed-forward neural networks (FNN), where, RNNs make use of the available sequential information rather than treating one time step as an isolated point. In RNNs, history is represented by neurons with recurrent connections, whereas in FNNs history is represented by context of $N - 1$. This is well presented in Fig. 1 where, the current state of hidden neurons c_t is obtained using the previous state of the network c_{t-1} and by applying weights w_r and w_{in} on the current input x_t . Based on this information, an output h_t is produced at each time step. In summary, RNNs are suitable in forecasting tasks where, the current output is not only based on the current input but also dependent on the information related to previous time steps. On the other hand, FNNs only use the presently available information to take a decision on the current output. In the context of LTLF, considering the information related to the current time step as well as the previous time steps are very crucial, i.e., a given monthly peak load value is not only related to the conditions of the current month, but also to the changes happened in previous months of previous years.

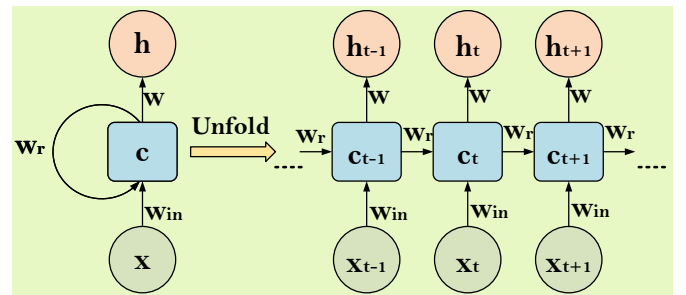


Fig. 1: Recurrent Neural Network Architecture

B. LSTM

LSTM can be introduced as an improved version RNN, where a gate mechanism is introduced to solve the problem of gradient explosion faced by traditional RNN. This sequence based deep learning architecture is capable of establishing temporal connections between the previous time steps and

the current time step [16]. LSTM architecture which is introduced in (1) to (6) represent how the concept of using different gates become useful in LTLF. This is shown in Fig. 2. In summary, LSTM network is designed to learn both the long-term and short-term features of the training data. This allows the algorithm to learn and memorise more important instances with short or long-term changes, resulting in a better prediction. It is important to understand the different utilisation patterns of the gates depending on changing input data. In other words how each gate learns the weight (W) and bias (b) respectively. In here, $\tanh$ (hyperbolic tangent) has been used as the activation function, where the activation function decides whether a neuron should be activated or not. Equation (5) represents the core of LSTM forecasting where, c_t stands for the cell state/internal memory at time step t and $\tilde{c}_t$ in (4) stands for candidate for cell state at time step t . Initially, cell memory c_{t-1} at time step $t - 1$ will be updated based on the input gate signal i_t in (1). In (1) it determines the output of current time step as per the previous state h_{t-1} and current input x_t at time step t . Similarly forget gate f_t in (2) updates the cell memory at previous time step based on the information irrelevant for the forecasting of next time step. Based on the updates, proportion of the results that should be executed as the output at time step t will be configured in output gate (3). Finally, the output h_t at time step t will be calculated in (6) based on the result from (3) and (5).

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (1)$$

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (2)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (3)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (4)$$

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t \quad (5)$$

$$h_t = o_t * \tanh(c_t) \quad (6)$$

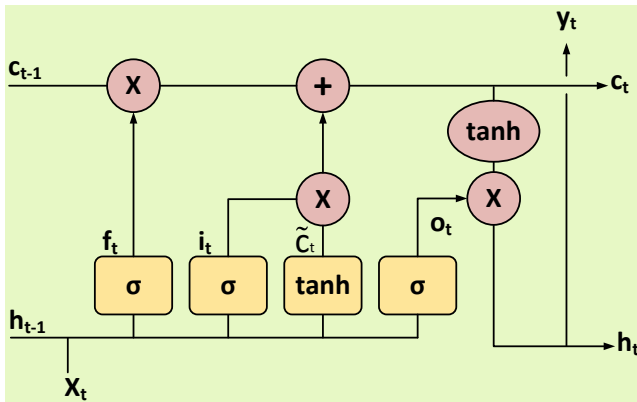


Fig. 2: LSTM Architecture

C. CNN

CNN, which was introduced by [20], can be defined as a DNN model which is mostly inspired by biological processes [21]. CNN Architecture have an input layer, an output layer

and one or more hidden layers. The input layer is the first layer that contains the input data. Convolutional layers are the core of CNN, which allow the neural network to study the features in each small local area rather than the entire data set. A pooling layer is usually added after several convolutional layers to reduce the size of the data as well as the number of variables, thus reducing the number of calculations needed to perform. Next, a fully connected layer is used to learn the features of the entire input rather than features of a local cluster and to produce the output.

In literature, CNN is widely used in image and video recognition, image classification, image processing, medical image analysis and natural language processing. Apart from these areas, CNN is also becoming popular in the area of time based flowing data [22], [23]. This is mainly because of its ability to process multichannel input data. CNNs use multiple building blocks such as convolution, pooling and fully connected layers to learn the spatial hierarchies of features. In the case of LTLF, researchers are dealing with multiple inputs and outputs, therefore, CNN can be introduced as a ideal selection for LTLF. However, when applied to numerical time-series data, CNN has poorer prediction accuracy compared to LSTM [18].

D. CNN-LSTM

LSTM is a quite well established technique that has performed better than other methods when it comes to time-series forecasting [24]–[26]. However, basic LSTM performs poorly in the application of latest available data for modelling long sequences and cycles in time-series [27], [28]. Compared to CNN, LSTM also has a weakness related to its capability of extracting the most valuable features from a given input. As already explained in the previous section, CNN has poorer prediction accuracy compared to LSTM. On the other hand, when applied to time-series classification, CNN has two main advantages over the other methods, i.e., its local dependency and scale invariance. Local dependency refers to the ability of CNN to correlate different signals, and scale invariance stands for the fixed scale for different paces or frequencies [29]. These advantages well fit with the targetted LTLF outcomes, where different factors as well as signals of different intervals affect the forecasted monthly peak load. Hence, the combination of CNN and LSTM contribute to increase the overall forecasting accuracy for time-series signals. Fig. 3, shows the overview of CNN-LSTM model that was developed in this work. A detailed explanation of the model is provided in the next section.

III. MODEL DEVELOPMENT

A. Data Pre-processing

The historical data for the past 260 months (i.e., 260 months spanning across January 1999 to August 2020) were obtained from the following public sources: historical temperature data for the state of NSW from Bureau of Meteorology (BoM) [30] and historical energy demand, social and economic data from Australian Energy Market Operator (AEMO) [31]. Initially, the

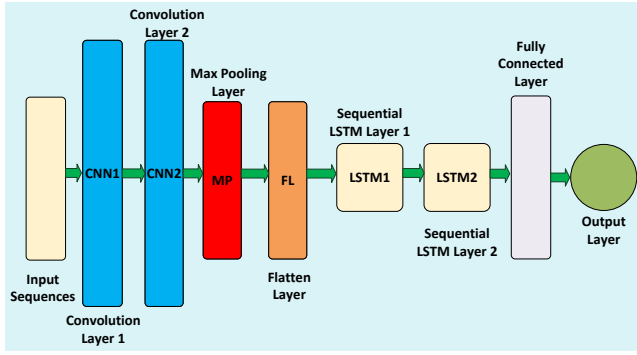


Fig. 3: CNN-LSTM Model.

target variable of total customer load was obtained for five-minute time intervals. Several related variables in different time intervals such as daily electricity price data, monthly maximum/minimum temperature data, quarterly economic and population changes, etc. were also obtained to the perform the LTLF task. A Matlab program was developed to convert these data that were sampled at different time intervals into a common interval of one month, e.g. calculating monthly peak load values from initial customer daily load profiles. In here, we have ensured that none of the data points are missed during the process of determining the monthly peak load value. This was achieved by picking the highest load value from the five-minute load values that were obtained for a period of one month (no data points were missing in the initial database for the calculation period). In some scenarios outliers maybe visible inside the data due to changes in system behaviour, fraudulent behaviour, human error, instrument error or simply through natural deviations. However, the customer load data that were used in this work has been accurately measured by AEMO, therefore, no outliers were visible in this instance. In addition, data related to over 90 other variables were also initially processed during the data pre-processing. For some of the original data that were sampled at an annual frequency, the Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) technique was employed in our program to interpolate the respective monthly values. As an example, the population changes per month have been interpolated based on the annual population changes. Apart from the readily available, few other user defined variables were also used to improve the training accuracy of sequential model. Next, Pearson correlation was performed between the target variable of monthly peak load and other related variables such as: monthly average load, mean value of the maximum temperatures, base population change, etc.

Table I summarises the variables that were highly correlated to the monthly peak load pattern. It includes ten variables identified through correlation analysis and another three variables identified through principal component analysis (PCA), which is discussed in the next section. Definitions of some of the variables are as follow: PV pattern represent the months which produce higher PV power generation compared to the other months of the year. This is due to the higher solar irradiance received during those months. In Australia, spring months produce the highest PV power generation. Peak load pattern

represents the months that result in maximum peak load values. In Australia, these are the summer months [2]. Participation rate is an estimate of an economy's active workforce. The load refers to the power in MW and consumption refers to the energy in MWh. Apart from these highly correlated variables, variables such as electric vehicle (EV) usage rate and distributed energy resource (DER) integration rate can also be considered in areas where DER integration or EV adoption is a major concern.

TABLE I: Applicability of Variables in the Proposed Method.

Variable Name	Application in PCA Necessity	
Peak Load	No	Essential
Average Load	No	Essential
Average Electricity Price	No	Essential
Maximum Electricity Price	No	Essential
Base Population Change	No	Essential
Participation	No	Essential
Consumption	No	Essential
PV Pattern	No	Essential
Peak Load Pattern	No	Essential
Incremental Peak Load	No	Essential
Mean Max Temperature	No	Essential
GSP Rate	Yes	Essential
Employment Rate	Yes	Essential
Population Rate	Yes	Essential
Monthly Lagging	Yes	Essential
Maximum Load	Yes	Essential
Unemployment Rate	Yes	Essential
Monthly Min/Max of	No	Optional
Daily Min/Max Temperatures	No	Optional
Electrical Vehicle Usage Rate	No	Optional
Renewable Energy Integration Rate	No	Optional
Dwelling Stock	No	Optional

B. PCA for Variable Aggregation

It is important to identify the features that are independent as well as the features that have dependency on each other (i.e., correlated with each other), therefore they can be aggregated as a whole. In long-term load forecasting, it is common to use more information in order to improve the accuracy of machine learning models, but the number of dimensions (n) increases as more features are added. As a result, forecasting results may over-fit due to the exponential increase in the number of configurations. Therefore, it is important to create a simple model that can run quickly as well as being easy to debug. Even though CNN architecture has the ability to extract the most relevant patterns from the input sequence, dimension reduction techniques reduce the overall complexity of the model by reducing the number of features in the input sequence that CNN has to go through in order to extract the most relevant patterns. For example, gross state production (GSP) growth and population growth along with the previous monthly peak load can affect the peak load pattern of the year, therefore, instead of representing this scenario by using three features, we can use two features. Dimensionality reduction techniques such as PCA, manifold learning, etc. are widely used, that would improve the model accuracy as well as

avoiding the over-fitting problems. PCA is identified to be more effective in dimension reduction compared to manifold learning [3].

PCA can be introduced as an orthogonal linear transformation technique that maps multiple variables into a new coordinate system such that all the features of original data are mapped. Initially, data were mostly dependent on each other, however, in the new coordinate system, these coordinates are orthogonal to each other. In here, the first coordinate, i.e. the first principal component, aligns with the greatest data variance, second principal component aligns with the second greatest data variance, and so on. The transformation can be defined as follows:

$$\begin{cases} T = XP \\ X^T X = PDP^{-1} \end{cases} \quad (7)$$

In here, X is the normalised mean-shifted data matrix with k feature columns where, each column is subtracted by the column mean. P can be introduced as a k -by- k matrix where, columns are eigenvectors of $X^T X$. D is the diagonal matrix with eigenvalues λ_j on the diagonal and zeros elsewhere. Finally, T is the new matrix developed, where, each column represents an independent feature projected to a principal component [3].

Explained Variance (EV) has been used to rank the independent features in order of importance. EV is a statistical measure of the amount of variability in a dataset that can be attributed to each individual principal component. After the conclusion of transformation, t independent features has been selected from the new data matrix T , which is shown below [3]:

$$EV = \sum_{i=1}^t \lambda_i / \sum_{j=1}^k \lambda_j \quad (8)$$

In this work, three independent features have been selected by transforming five economical-population-load features along three orthogonal principal components. The corresponding EV value was calculated to be 97.9%. In other words, majority of the important information embedded inside the original five features have been brought forward in the newly constructed features $PCA1$, $PCA2$ and $PCA3$.

C. Sequence to Sequence Modelling

A seq2seq deep learning model has been incorporated in this work to construct an ensemble multivariate time series model that can predict the monthly peak load changes for a three year period. A model can be defined as a seq2seq model if the inputs as well as the outputs follow a sequence. It is quite well known that network architectures have few different sequence configurations that being used in sequence prediction problems [15]. They can be introduced as: one-to-one, one-to-many, many-to-one and many-to-many.

The focus of this research has been on forecasting the monthly peak load for the future three years (i.e. future 36 months) taking into consideration of the previous three years data. Therefore, many-to-many configuration matches with the research target. Fig 4 demonstrates the recommended many-to-many deep learning structure, where past 36 months' (i.e., past

three years) peak load data and other factors are considered to forecast for the future 36 months' (i.e., future three years) peak load. A typical seq2seq network model consists of two parts; a) an encoder and b) a decoder. In here, CNN network architecture works as the encoder to understand the input sequence and to create a smaller dimensional representation of it. If forecasting of 9/2017-08/2020 is taken as an example, inputs to the encoder will be: peak load, average load, average price, maximum price, base population change, participation, consumption, PV pattern, peak load pattern, incremental peak load, mean max temperature, $PCA1$, $PCA2$ and $PCA3$ of the immediate past months 1/2015-8/2017. The outcome of the encoder will be next forwarded to the decoder, which is the LSTM network. The outcome of encoder will have a smaller dimensional representation of initial input data along with reduced number of variables. The decoder will next generate its own representation of the output sequence. After one whole iteration, an output sequence of future three years will be generated based on the submitted information related to the past three years.

IV. MODEL TRAINING, VALIDATION AND TESTING

The training, validation and testing consist of multiple steps, which are; isolation of most recent three years data to evaluate the model performance, data sequential arrangement, feature scaling, splitting the remaining data into training and validation, model training and parameter optimisation. The following discussion applies to the process flow chart shown in Fig. 5.

A. Recent Data Isolation

In a sequential model, training and validation data will be randomly split, therefore, the data related to the evaluation of most recent 36 months (i.e., 09/2017 to 08/2020) were isolated before feeding the rest into the model. This ensured that there is no data leakage between the target test evaluation dataset and the data used in the initial model training and validation.

B. Data Sequential Arrangement

Lack of data availability is one of the major drawbacks that can degrade the training accuracy of LTLF models. In order to obtain highly accurate forecasting results, a LTLF model requires much more data than a STLTF model. Sequential arrangement of the data is a technique that allows to overcome this.

In this work, 153 different sequences of raw data were obtained for the purpose of training and validation, where an individual sequence consists of data related to 72 months. Out of the 72 months, first 36 months will be the "Train/Validation-X set" and last 36 months will be the "Train/Validation-Y set", such that data of previous 36 months will be used to predict the future 36 months' monthly peak demand. "Train/Validation-X set" will have historical data related to 14 different variables: "Monthly Peak Load, Average Load, Maximum Electricity Price, Population Change, etc". On the other hand, "Train/Validation-Y set" will be the target or prediction variable, "Monthly Peak Load".

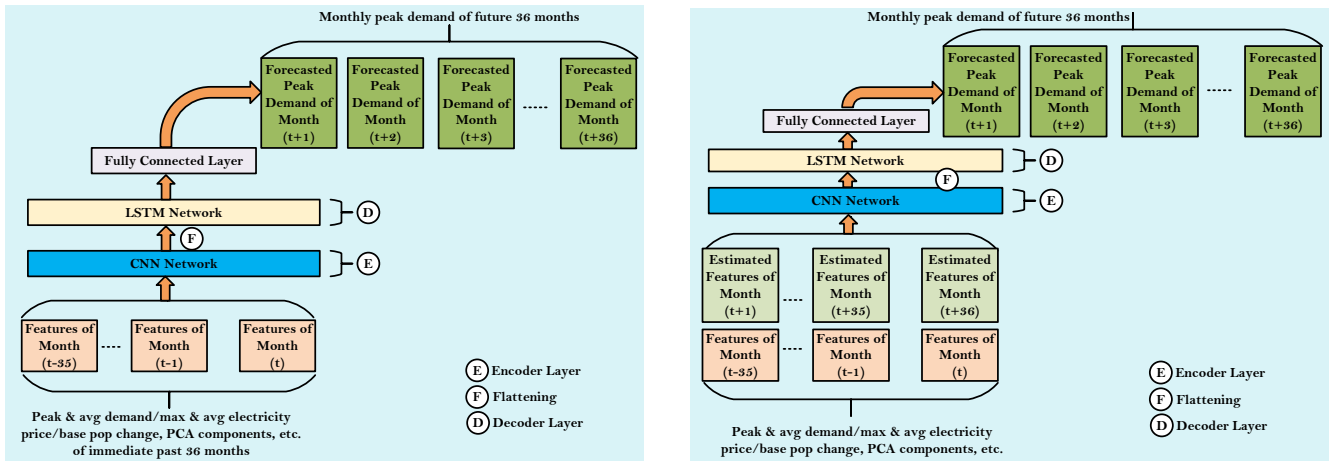


Fig. 4: Sequence to Sequence Modelling of Test Data a) Using Historical Data b) Using Historical and Estimated Future Data

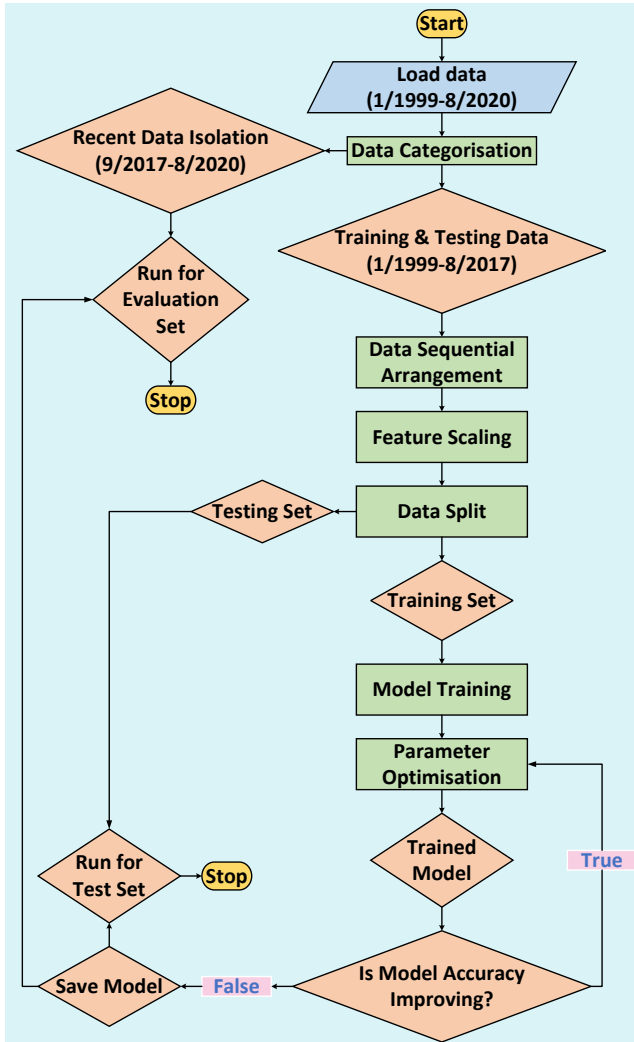


Fig. 5: Process Flowchart.

Suppose “Train/Validation” split ratio is 90%, it is common to use the first 90% of the 153 sequences as the “Train set” and the latter 10% as the “Validation set”. However, in this work “Train set” and “Validation set” sequences were picked randomly by ensuring that, if a sequence is used as a

“Validation set”, then it will not again appear inside the “Train set” and vice versa. This resulted in 137 and 16 sequences as the “Train set” and “Validation set” sequences, respectively. This randomised approach increased the model training accuracy and helped to obtain a highly accurate LTLF model. Table II shows an example that is structured according to the many-to-many sequential configuration, where 1/1999 up to 12/2001 were the training months of the model. This sequence predicted the next 36 months from 1/2002 up to 12/2004. Similarly, another 152 different records were generated using the available monthly data from 1/1999 to 8/2017. After the model was trained for one round, the forecasted values were compared against the actual values to calculate the forecasting accuracy of the model.

C. Feature Scaling

Scaling the different features into a common ground is a key data pre-processing step in any given machine learning algorithm. If the data is not properly scaled, algorithm may bias towards numerically large values. In this study, two most widely used feature scaling techniques were tested: a) standardisation b) min-max scaling. Generally, standardisation is useful in scenarios where the data follows a Gaussian distribution. Out of the two techniques, it was identified that standardisation fits well with the task of LTLF.

$$Z = (X - \mu) / \sigma \quad (9)$$

Where, X is the original value of a given variable, μ and σ stands for the mean and the standard deviation of a given variable. The forecasted peak load values will be de-standardised before comparing with the actual values.

D. Data Split

Typical data split ratio in a machine learning algorithms is 80%/20% [32]. In this study, different data split ratios were tested and it was identified that 90%/10% resulted in the best outcome. The 153 different records were randomly split between the training data and validation data. This introduced random variations into the monthly peak load patterns and resulted in improved training accuracy.

TABLE II: Dataset Example for Many-to-Many Configuration.

Month ID	Training Sequence	Historical Training Features													Testing Sequence	
		Peak Load	Avg Load	Avg Price	Max Price	Base Population Change	Participation	Consumption	PV Pattern	Peak Load Pattern	Incremental Peak Load	Mean Max Temperature	PCA1	PCA2		PCA3
1	Jan 1999	10002.84	7389.48	23.22	333.26	0	62.01	48171.74	0	1	271.69	26.6	17867.85	-821.23	41.55	Jan 2002
2	Feb 1999	10274.53	7451.92	19.83	133.84	5.55	61.97	48439.00	0	1	-703.05	26.1	17597.80	-814.63	38.35	Feb 2002
3	Mar 1999	9571.48	7446.53	17.52	57.3	11.63	61.91	48655.81	0	0	304.57	25.7	17512.63	-545.58	48.60	Mar 2002
....																
34	Oct 2001	9212.00	7593.24	27.51	120.62	210.98	62.64	52449.13	1	0	620.39	24	11764.15	-896.10	48.38	Oct 2004
35	Nov 2001	9832.39	7676.91	21.72	52.68	215.27	62.57	52555.00	1	0	673.09	23.5	11402.83	-1796.77	18.68	Nov 2004
36	Dec 2001	10505.48	7586.12	29.69	1766.45	219.88	62.55	52687.44	0	1	476.13	26.2	11192.26	-1182.68	29.86	Dec 2004

E. Model Training and Parameter Optimisation

The intention of this section is to provide a brief overview of the CNN-LSTM model shown in Fig. 3 and how the network parameters were optimised to improve the forecasting accuracy. After the pre-processing of data, Train-X data were passed through to two convolutional layers, where each of them consisted of 64 and 36 filters, respectively with a stride value of three each. In here, filter represents the number of output channels after convolution has been performed on the input sequence. On the other hand, kernel represents the size of a convolution filter being used to perform the convolution on the input sequence. Upon applying convolution on the input sequence for a single filter, we slide a 3x3 matrix (i.e., kernel stride value) across the input sequence that gives us a single channel output.

The extracted input data features are then sequentially passed through to the max pooling layer. Pooling layer is a layer added after the convolutional layer and it is the last element in a CNN architecture. Its function is to progressively reduce the spatial size of the representation to reduce the network complexity and computational cost. Pooling involves selecting a pooling operation, this is much like to the filtering done in convolution layer. The size of pooling operation is smaller than filtering happened in the previous layer. It is most of the time set with a stride value of 2x2, such that a 2x2 matrix is slide across the output from previous layer. This means that the pooling layer will always reduce the size of each feature map by a factor of 2. Max pooling is simply a rule to take the maximum of a region and it helps to proceed with the most important features from the output sequence from convolution. In brief, pooling layer creates summarised version of the features detected at the CNN layers.

Next, all multidimensional arrays from pooled feature maps will be converted into a single long continuous vector by sending through a flatten layer. Next, the output from the flatten layer is sent through a single repeat vector layer which prepares a 2D array input for LSTM layers in the decoder. The decoder layer is designed to unfold the the encoding, therefore, they are stacked in the reverse order of the encoder. Layer 6, i.e., LSTM-64, reads the output data from the encoder and provides an output of 64 features with 36 time-steps for each, since the return sequence is set to be true in this layer. Next, layer 7, i.e., LSTM-36, reduces the feature size to 36 with 36 time-steps for each. Finally, the TimeDistributed layer, i.e.,

Dense-36, is added at the end to obtain the output from the network, where it creates a vector of length 36, which is equal to the number of features outputted from the previous LSTM layer. Relu activation function was used in both CNN and LSTM layers, considering that Relu converges more quickly and reliably than other deep neural network training activation functions like sigmoid, Tanh, Leaky Relu, etc. At this stage, prediction training process is completed.

Next, it continues with the assessing process where error of the predicted values will be calculated. In other words, the predicted value is compared with the actual value to obtain the forecasting error. These results perform as the reference to determine whether the stopping conditions of the network are met. In this scenario, the training stop condition was determined to be a EarlyStopping with a patience of 15 instances. If the error of forecasting does not improve continuously for 15 occurrences, then the network will automatically stop the training and the finalised network configuration will be saved. If the stop condition is not met, the calculated error will be back-propagated to the previous layer such that weights and bias values at each layer will be adjusted. Then, repeat the training process starting from the first layer. Adam optimiser along with the loss functions Mean Absolute Error (MAE) and Mean Squared Error (MSE) were used in the compilation process.

A Bayesian optimisation code was developed to optimise the network hyper-parameters. In this work, python library of HyperOpt for sequential mode based optimisation [33] was used to perform the Bayesian optimisation. Table III summarises the optimised network parameters that resulted in the best outcome of the network.

V. DISCUSSION ON POINT FORECASTING RESULTS

A. Experiment Procedure

Fig. 6 demonstrates four of the sixteen different test case scenarios obtained using the proposed CNN-LSTM model during the training and validation. These represent the worst case scenario, two normal scenarios and the best case scenario. These were obtained for the periods of 5/2017-4/2017, 12/2008-11/2011, 5/2003-4/2006 and 8/2004-7/2007 respectively, such that it covers the whole period. MAPE values of 5.73%, 4.64%, 3.6% and 3.34% were obtained respectively. According to the results, it can be seen that the model was able

TABLE III: Optimised Hyper-parameters

Layer	Hyper-parameter	Range	Optimal Value
CNN	Convolutional layer 1 filters	1-128	64
CNN	Kernel stride value convolutional layer 1	1-5	3
CNN	Convolutional layer 2 filters	1-128	36
CNN	Kernel stride value convolutional layer 2	1-5	3
CNN	MaxPool layer pool size	2	2
CNN	Convolutional layer 1 and 2 activation function	Relu	Relu
LSTM	No of units in LSTM layer 1	1-128	64
LSTM	No of units in LSTM layer 2	1-128	36
LSTM	LSTM layer 1 and 2 activation function	Relu	Relu
FC	No of units TimeDistributed dense layer 1	36	36
FC	No of units TimeDistributed dense layer 2	36	36
Model Fitting	Optimizer	Adam	Adam
Model Fitting	Loss Function	MAE and MSE	MAE and MSE
Model Fitting	No of epochs	10-1000	220
Model Fitting	Batch Size	1-72	8

to achieve good results for each case scenario while achieving a best MAPE of 3.34% and a worst MAPE of 5.73%.

After the initial training and validation of the proposed CNN-LSTM model using the predicted data vector, the model was used to evaluate the monthly peak load of the target data set, which was 36 months starting from 9/2017 to 8/2020. As previously stated, the target months were isolated before the model training phase to ensure that model is capable of predicting the future which can slightly deviate away from the general trend in the past. Following factors were taken into consideration when the target period was selected out of the available data.

- 1) It was ensured that both the summer months and winter months were captured as a whole in the target data set.
- 2) The selected period demonstrated significantly higher summer peak loads compared to the winter peak loads in the same horizon.
- 3) It was noticed that part of the selected period is affected by the economical and behavioural changes that were caused by the Covid-19 pandemic.

Two approaches can be used to estimate the economic, population and climate features of the target period (i.e., future years); a) Next few years' forecasted features can be obtained from the third-party or government organisations, b) Features belonging to the immediate past years can be modified to obtain a futuristic representation of the next couple of years. In this work, the latter approach was used in the evaluation process. It should be noted that both these methods may contain some level of error. However, these numbers are accurate enough for the forecasting of immediate coming years. Fig. 4-a shows how CNN-LSTM network was unrolled for the testing period using historical economic, population and climate data of past 3 years and Fig. 4-b shows how CNN-LSTM network was unrolled for the testing period using estimated economic, population and climate features of the target period along with the immediate past months monthly peak load values. Testing result discussion is based on the

latter approach considering the common industry practice of it as well as improved forecasting results obtained.

Fig. 7 demonstrates the forecasted results obtained for the target period using the developed CNN-LSTM model. In order to confirm the improved accuracy of the model and also the applicability of CNN-LSTM technique in LTLF, the obtained results have been compared against separate results obtained using a CNN network model and a LSTM network model. To calibrate the models, the statistical metrics MAE and MAPE were computed to show the model accuracy. MAE is represented by (10) where the units are MWs and MAPE can be represented by (11).

$$MAE = \frac{1}{N} \sum_{i=1}^N |ForecastedLoad_i - Trueload_i| \quad (10)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \frac{|ForecastedLoad_i - Trueload_i|}{|Trueload_i|} \quad (11)$$

A desktop PC with a 10th generation Intel core-i9 processor with 16GB RAM was used to run the algorithms along with software configurations of Windows 10 and Python 3.8 with Tensorflow backend.

B. Results Analysis

Table IV below demonstrates the results obtained using the three models for the period of 36 months. CNN-LSTM model obtained the best MAE/MAPE value of 499.85MW/4.29%, while the CNN and LSTM models achieved a MAE/MAPE value of 523.83MW/4.55% and 583.76MW/4.91% respectively for the target 36 months period.

The following observations were drawn out of the obtained results:

- 1) Overall, the proposed CNN-LSTM model performed better than other two models when applied to LTLF of 36 months ahead. This is quite visible where CNN-LSTM achieved the largest MAPE of just 13.63% during the forecasted 36 months. In comparison, CNN and LSTM models performed worse than the CNN-LSTM model, where highest MAPEs of 14.29% and 15.27% were obtained, respectively. Also, looking at the average values of all MAPEs that have exceeded the 10% threshold indicate that, CNN-LSTM, CNN and LSTM models have resulted in 11.48%, 11.89% and 12.38%, respectively. This again proves the dominance of CNN-LSTM in LTLF.

It should be noted that the forecasting of future 36 months was performed in a single iteration, whereas in other methods, the firstly forecasted 12 months will be fed as an input into the model to forecast the subsequent 12 months [3], which is consuming more time and resources. This proves the applicability of CNN-LSTM in LTLF up to 36 months ahead.

- 2) Observing the forecasting results of spring-summer months in Australia shows that proposed CNN-LSTM model is well ahead of the other two models. In summary, for spring-summer CNN-LSTM achieved an overall MAPE value of 5.30% while the other two methods achieved 5.83% and 5.93% respectively.

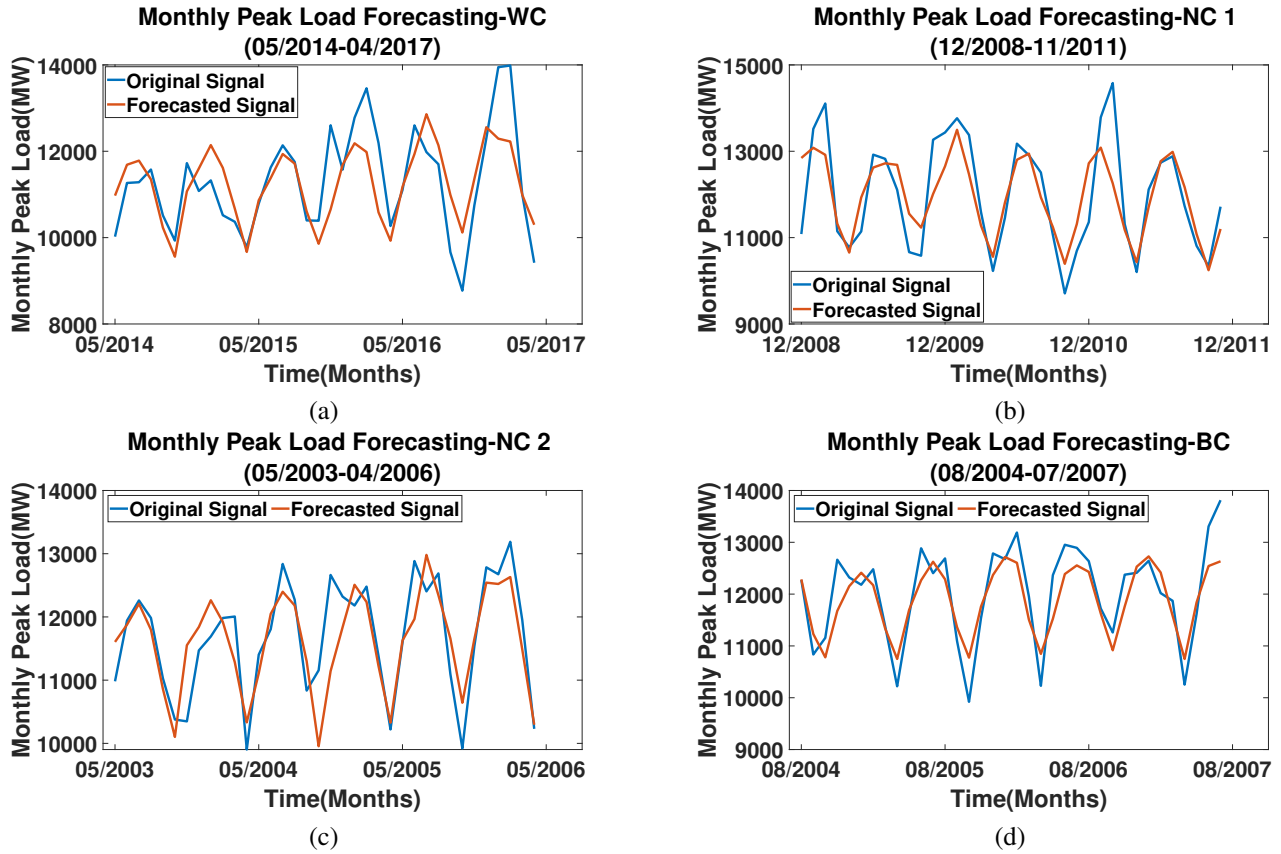


Fig. 6: Forecast results of model validation (a) Worst Case Scenario (b) Normal Scenario 1 (c) Normal Scenario 2 and (d) Best Case Scenario.

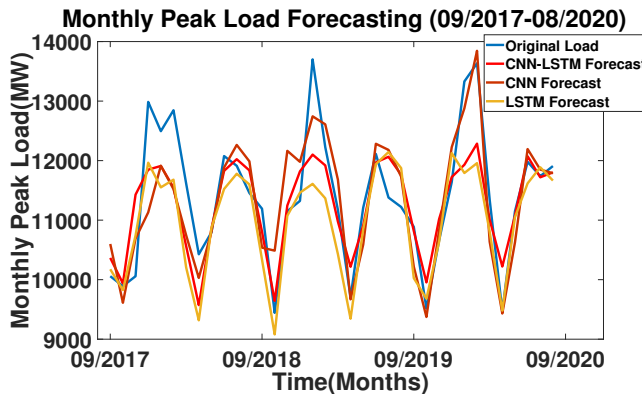


Fig. 7: Comparative Forecasting Results of Algorithm Evaluation.

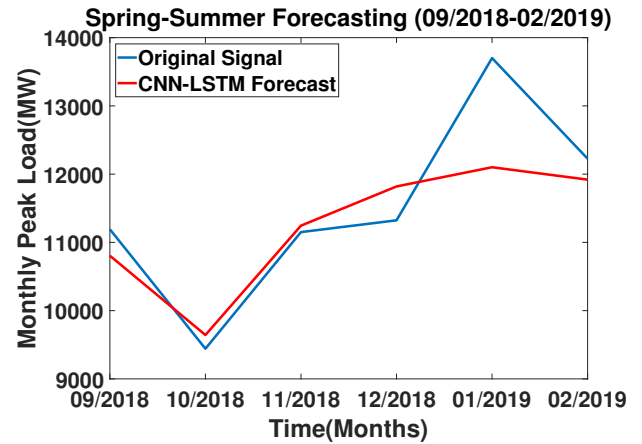


Fig. 8: Spring-summer Monthly Load Evaluation.

In Australia, spring-summer months experience considerably less rain without a clear visible pattern compared to the autumn-winter months as per the statistics from BoM [30]. This unrecognisable rain pattern results in larger fluctuations in the temperature during this period. As a result, customer load patterns also experience higher fluctuations compared to the winter months. This is mainly because of the fluctuations happening in the usage pattern of electric air conditioners. WP has recognised that having 3-4 days of continuously high temperature will result in higher usage of air conditioners. However,

with temperature patterns continuously changing due to rainy days happening in random intervals, the usage pattern of air conditioners also varies. This is the main reason why spring-summer LTLF prediction values result in a higher MAPE value than the autumn-winter months.

Compared to the other two methods, CNN-LSTM model has several main advantages such as: ability of CNN to a) correlate different signals b) scale-invariant for different paces or frequencies; and ability of LSTM to a) well predict the short and long-term dependencies in a time-series b) improved accuracy in predictions.

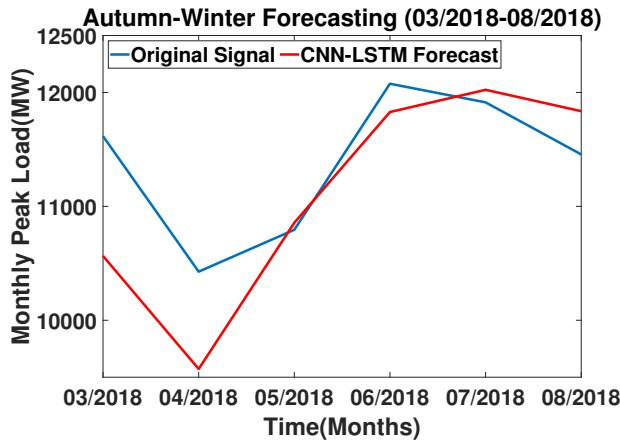


Fig. 9: Autumn-winter Monthly Load Evaluation.

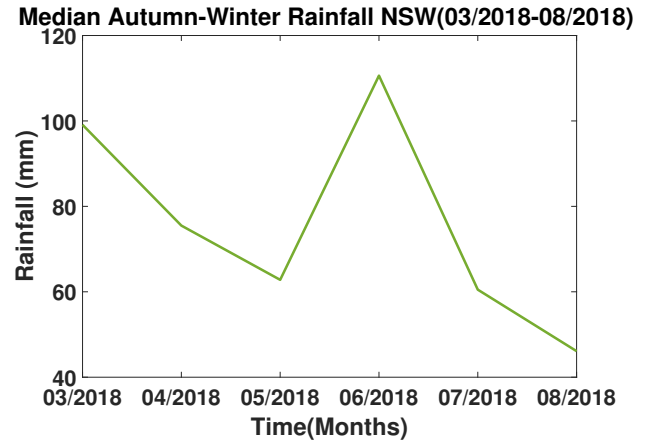


Fig. 11: Autumn-winter Monthly Median Rainfall.

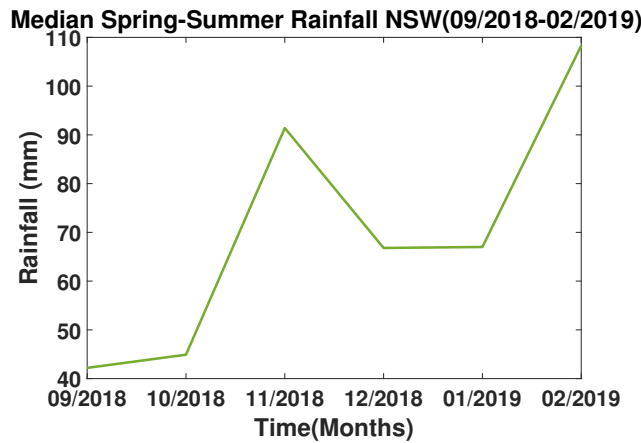


Fig. 10: Spring-summer Monthly Median Rainfall.

Therefore, collectively this method resulted in a better prediction accuracy in spring-summer months.

- 3) Compared to the spring-summer months autumn-winter months experience less fluctuations and lower customer demand. This is mainly because of the continuous raining pattern happening in Australia during these months. This results in relatively low and more predictable fluctuations in daily temperatures. On the other hand, compared to the spring-summer, autumn-winter use more gas for heating purposes, this is the main reason why there is less customer demand compared to the other half of the year. Both CNN-LSTM and CNN models achieved similar MAPE values in this period (i.e., 3.29% and 3.28%). This is improving the prediction accuracy by around 50% compared to the spring-summer. Most importantly, this outcome agrees with our explanation earlier, “compared to spring-summer months autumn-winter months experience less fluctuations”. This nature makes the prediction process less complicated, therefore, both methods achieved good prediction outcomes.
- 4) This outcome is matching with the seasonal load analysis results shown in Fig. 8 and Fig. 9, as well as with the seasonal rain pattern analysis data shown in Fig. 10 and Fig. 11. A statistical analysis of spring-summer and autumn-winter load data was performed to verify the pre-

vious outcome. Maximum load deviation (standard load deviation) of 0.31pu (0.09pu) and 0.14pu (0.04pu) were obtained for spring-summer and autumn-winter months, respectively. On the other hand, maximum rainfall deviation (standard rainfall deviation) of 0.61pu (0.22pu) and 0.58pu (0.20pu) were obtained for spring-summer and autumn-winter months, respectively. Based on this analysis, it can be said that there is a positive relationship between the variations happening in monthly customer load data and variations in monthly rain pattern. This is the reason why spring-summer customer load values are embedded with larger fluctuations than autumn-winter. As a result of this, predicting the spring-summer customer load becomes lot harder than predicting the autumn-winter customer load.

- 5) This study overcomes the drawbacks of current researches by forecasting the whole year at once, whereas, many currently published work forecast the winter months separately and summer months separately to avoid the larger fluctuations that may happen in demand due to seasonal changes or to avoid the months that affect considerably less towards the yearly peak load [3], [13], [34].
- 6) In this work, a point forecasting method has been used to forecast the monthly peak load for a forecasting horizon of three years. In majority of work done on monthly peak load point forecasting, the maximum horizon has been a year ahead. Nevertheless, having an accurate load forecasting for a horizon of more than 1 year allows the network operators to carry out their planning activities with a great confidence. Therefore, three years ahead LTLF provides the necessary understanding of the future power system to the planning services.

VI. IMPLEMENTATION OF PROBABILISTIC FORECASTING FRAMEWORK

Forecasting by nature can be interpreted as a stochastic problem, where long-term forecasting consists of multiple stochastic features. Even though point forecasting model provides accurate results for the target period, this has not taken into account the stochastic nature of long-term forecasting. Mainly related to the uncertainties associated with long-term

TABLE IV: Three Years Ahead Forecasting Results.

Month	CNN-LSTM		CNN		LSTM	
	MAE(MW)	MAPE(%)	MAE(MW)	MAPE(%)	MAE(MW)	MAPE(%)
9/2017	307.84	3.06	541.53	5.38	116.79	1.16
10/2017	45.34	0.46	280.24	2.83	65.86	0.67
11/2017	1370.86	13.63	633.16	6.30	684.09	6.80
12/2017	1137.63	8.76	1856.04	14.29	1020.85	7.86
01/2018	584.51	4.68	580.38	4.65	944.12	7.56
02/2018	1297.37	10.10	1324.84	10.31	1166.24	9.08
03/2018	1051.40	9.05	869.74	7.49	1410.16	12.14
04/2018	854.19	8.19	397.49	3.81	1109.75	10.64
05/2018	61.76	0.57	7.53	0.07	76.59	0.71
06/2018	248.36	2.06	191.35	1.58	553.17	4.58
07/2018	110.31	0.93	350.58	2.94	134.13	1.13
08/2018	379.62	3.31	531.87	4.64	144.04	1.26
09/2018	386.98	3.46	653.93	5.84	856.69	7.65
10/2018	200.61	2.12	1045.35	11.07	362.01	3.83
11/2018	94.52	0.85	1013.27	9.09	76.75	0.69
12/2018	496.23	4.38	656.19	5.80	138.08	1.22
01/2019	1599.70	11.68	956.59	6.98	2091.64	15.27
02/2019	306.68	2.51	383.15	3.13	859.30	7.03
03/2019	197.72	1.77	506.51	4.53	744.22	6.66
04/2019	513.29	5.29	34.32	0.35	359.45	3.70
05/2019	300.17	2.68	619.78	5.53	361.03	3.22
06/2019	143.40	1.18	170.17	1.40	173.06	1.43
07/2019	684.81	6.02	800.29	7.03	763.98	6.71
08/2019	540.50	4.82	509.17	4.54	653.50	5.82
09/2019	69.68	0.64	668.51	6.14	861.07	7.91
10/2019	438.99	4.61	141.92	1.49	165.21	1.74
11/2019	360.63	3.38	172.35	1.62	20.22	0.19
12/2019	84.61	0.73	586.75	5.04	487.75	4.19
01/2020	1399.50	10.50	451.32	3.39	1538.40	11.54
02/2020	1349.48	9.90	210.21	1.54	1676.81	12.30
03/2020	335.42	2.96	706.35	6.23	539.78	4.76
04/2020	749.55	7.91	39.76	0.42	8.66	0.09
05/2020	74.73	0.67	522.52	4.69	97.75	0.88
06/2020	93.01	0.78	213.99	1.79	361.82	3.02
07/2020	19.16	0.16	110.62	0.94	149.17	1.27
08/2020	105.93	0.89	119.95	1.01	243.38	2.04
Mean	499.85	4.29	523.83	4.55	583.76	4.91

prediction of economic, population and climate changes, it is always encouraged to obtain a probabilistic forecasting that combines multiple forecasting results for a given period. This section discusses how a probabilistic forecasting approach was developed for the target period of 9/2017 to 8/2020.

A. Scenario Generation

In this work, a standard block bootstrap method [35] was used to generate different scenarios for the multiple years ahead forecasting process starting from year t (9/2017) up to year $t + 2$ (8/2020). In here, block bootstrap method is used as a tool for scenario generation by considering economic, population as well as climate data, such that we can improve the probabilistic forecasting accuracy of the CNN-LSTM model. The following three criteria were carefully determined before the actual scenario implementation.

- 1) Selection of different factors for the scenario analysis: In here, all variables that were used in the process of point forecasting were used to develop the model and some of the key indicators were used as the inputs to generate different scenarios, such as, historical mean value of the maximum temperatures, monthly average load, monthly average price, employment and GSP. In other words, historical values act as the future values of the input variables that are required for the forecasting.
- 2) Method implementation and historical year selection: A block bootstrapping method was used in this study, where a standard method for bootstrapping a time-series involves taking random segments of the historical time-

series data and pasting them together to form a new artificial series. In this way, a large number of different series can be formulated, however, length of each block/segment should be selected such that it is long enough to capture all the required information and short enough to generate a large number of simulated series.

The number of historical years to be selected for scenario analysis varies with different forecasting approaches. As an example, [13] used quarterly temperature data related to 27 years from 1980-2007 to simulate different temperature series. In this work, data related to the past 18.5 years as well as the simulated data related to the forecasting period were also used to generate multiple bootstrap series. This was done by modifying the data related to the most recent past to match the trend of the actual data of 9/2017 - 8/2020, such that there won't be any representation of the actual testing data inside the model training. This allowed to introduce more data into out bootstrap method, such that it provides a large range of possible bootstrapped series. The series show some serial correlation up to about three months based on the same seasonal pattern. Therefore, the size of a block was designed to be three months, such that it covers a period of one season (i.e., spring, summer, autumn or winter). One bootstrapped series consists of 12 of these blocks to represent a period of 36 months. In this way, data for each season stays together but the same season from different years get rearranged, such that the bootstrapped series is a shuffled version of the original series with the same length. For example, some of the generated bootstraps may have the structure shown in Fig. 12.

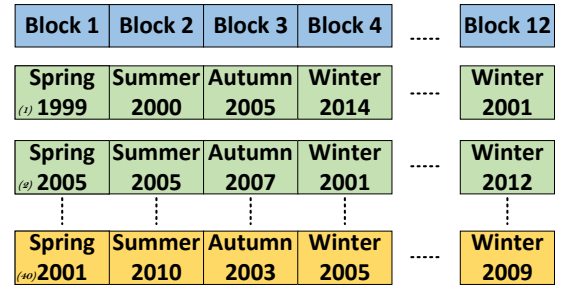


Fig. 12: Bootstrapped Series.

According to the above example, historical economic, population and climate data related to a period of 36 months (i.e., spring-1999, summer-2000, winter-2001 etc. shuffled in blocks of 3 months) will be used as a time-series to forecast the target period of 9/2017 to 8/2020. In other words, we assume that spring of 2017 can be similar to conditions back in 1999, summer of 2017 can be similar to summer of 2000, winter of 2020 can be similar to winter of 2001, etc. These randomly selected years gave rise to a large range of possible bootstrapped series where, 40 different scenarios were obtained. For each of this scenario a load profile was generated for the target period of 36 months.

- 3) Normalised monthly peak load: Each of the 40 scenarios generated a monthly peak load profile for the

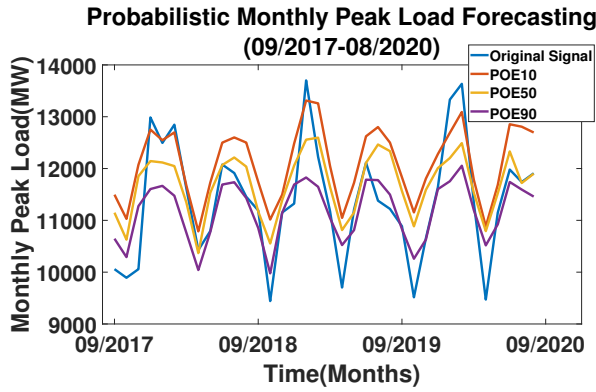


Fig. 13: Monthly Peak Load Normalisation (9/2017-8/2020).

period of 9/2017 to 8/2020. These individual load profiles were used to obtain the probabilistic monthly peak load forecasting results. The normalised monthly peak load or the 50th percentile is a key parameter that can be obtained using these results. Usually, POE values of 10% (90th percentile), 50% (50th percentile) and 90% (10th percentile) are calculated to support the decision making.

B. Probabilistic Forecasting

Fig. 13 shows the obtained probabilistic forecasting results for the target period where, POE10 (orange), POE50 (yellow) and POE90 (purple) curves are plotted on top of the actual monthly peak load values labelled in blue colour. The POE10 is usually used in practice to represent a severe scenario that may happen just 10 times out of 100 times. This can be applied to this study as well.

According to the explanation given under section V-A, the selected evaluation period experienced significantly higher summer peak loads compared to the historical years that were used in the training process. This is shown in Fig. 13 where few of the actual peak load values are above the POE10 curve, however, other than this extreme scenario, the POE10 curve was managed to capture the general shape of the monthly peak load curve. This implies that POE10 curve derived from 40 different scenarios is not sensitive to the extreme value, however, it is still reliable and practical to use the POE10 curve for planning purposes.

The overall probabilistic forecasting accuracy is evaluated via continuous ranked probability score (CRPS). CRPS is calculated by comparing the cumulative distribution functions (CDF) for the forecast against a reference dataset (observations, or analyses) over a given period, where CRPS does not focus on any specific point of the probability distribution but considers the distribution of the forecasts as a whole [36]. CRPS for each month across the forecasting horizon of 36 months (9/2017 to 8/2020) was calculated by comparing the CDFs of the 40 different scenarios against the actual monthly peak load. The average CRPS of three years was calculated to be 509.15 MW and as a percentage 4.64%. This agrees with the visual probabilistic predictions shown in Fig. 13. In order to highlight the applicability of block bootstrapping in LTLF, we have compared our results with probabilistic

LTLF results obtained using a quantile forecasting model (i.e., pinball loss function). In here, the average CRPS of three years was calculated to be 536.87 MW and as a percentage 4.66%, which is worse than our method. This proves the applicability and accuracy of block bootstrapping in LTLF.

C. Key Highlights of the Model

- 1) The proposed CNN-LSTM model accuracy was improved by introducing time distributed layers to the fully connected layers and keeping the return sequence of decoder equal to “True”. This was done to avoid flattening the neural network output and to keep each time-step value separate for the future 36 months. This prevented the model from having random interactions between different time-steps and only allowed interactions between the values of its own time-step.
- 2) Apart from the traditional inputs that are used for LTLF, the authors introduced the yearly rooftop photovoltaic (PV) power generation pattern as a separate training input. This was informed by the practical industry experience from WP. According to WP, in Australia, the spring months produce relatively higher PV power generation compared to the other months. In our model, this factor was introduced as a separate input, which resulted in better prediction accuracy compared to the other two models during the spring-summer months.

D. Results Comparison

Table V compares the outcome of this work against the currently published work, MAPE value is the only parameter compared here considering the size of different energy markets in these studies. Also, in places where prediction has been done separately for summer and winter, the average MAPE of those two was taken to make a logical comparison. Please note that most of the published work does not match with the time horizon of 36 months ahead. To avoid any ambiguity, two of the most recently implemented long-term load forecasting models (i.e., Gated Recurrent Units (GRU) model and Support Vector Regression (SVR) model) in the literature [3], [9] have been tested with the same data that were used in this work. Even though, these models obtained interesting results in their respective research work, they performed poorly in long-term three years ahead monthly peak load forecasting.

The results prove that proposed point forecasting method has outperformed all other methods, even performing better than one year ahead forecasting methods. As per the results, it can be concluded that the model is capable of forecasting the load demand for the next three years with a very high accuracy of 95.71%.

On the other hand, the following highlights can be drawn out with respect to the probabilistic forecasting approach. Method suggested by [12] has achieved a MAPE value of 3.83% in four years ahead probabilistic forecasting. Nevertheless, the work scope has been limited to substation level, whereas in our work the focus is on the grid level. Also using a time interval of one hour in [12] does not add a significant value towards the long-term decisions as it was mentioned under section I. In

TABLE V: Results Comparison with Existing Work.

Method	MAPE	Time Interval	Time Horizon	Point or Probabilistic Forecast	Reference
Density Forecasting	N/A	Weekly Yearly	Ten years ahead	Probabilistic	[13]
LSTM	6.54	Hourly	Five years ahead	Point	[8]
CNN-LSTM	5.12	Monthly	Five years ahead	Point	N/A
Point Process Model	3.83	Yearly	Four years ahead	Probabilistic	[12]
CNN-LSTM	4.29	Monthly	Three years ahead	Point	N/A
CNN-LSTM	4.64	Monthly	Three years ahead	Probabilistic	N/A
GRU	6.02	Monthly	Three years ahead	Point	N/A
SVR	16.94	Monthly	Three years ahead	Point	N/A
Fuzzy-ANN	5.73	Monthly	Three years ahead	Point	[10]
Fuzzy Logic	6.9	Monthly	One year ahead	Point	[7]
Model L	4.7	Monthly	One year ahead	Probabilistic	[14]
LSTM/GRU	5.815	Yearly	One year ahead	Point	[3]
ARIMA	12.92	Yearly	One year ahead	Point	[3]
LLE-LSTM	6.45	Monthly	One month ahead	Point	[16]

[14], it has achieved a MAPE value of 4.7% for a forecasting horizon of one year, whereas in our probabilistic forecasting method it is aimed to forecast for a period of three years. The proposed probabilistic forecasting approach has achieved a forecasting accuracy of 4.64%, which is still better than most of the published work. Nevertheless, its accuracy can still be improved by taking into account other factors that may have a significant influence towards scenario generation. This will be considered in our future work.

Overall, it can be said, the actual performance of different forecasting techniques hugely depends on the forecasting time horizon that it is being applied to. CNN-LSTM method introduced in this work has performed very accurately when it comes to long-term tactical planning (1-3 years ahead planning) and even being able to extend the forecasting horizon up to 5 years ahead with a minimal effect on the overall accuracy. Also, it is fair to say that this method has the ability to perform with the same high accuracy when it comes to STLF, MTLF and general long-term tactical planning (few weeks to few months ahead). Methods such as GRU, fuzzy-ANN, fuzzy logic and LLE-LSTM have also achieved interesting results for long-term forecasting horizons up to 3 years. Nonetheless, their accuracy is lower than the proposed method. On the other hand, it is reasonable to say that probabilistic forecasting methods perform better than point forecasting techniques when it comes to strategic long-term planning (5/10/15/20 years ahead planning). Methods such as, CNN-LSTM, point process model, density forecasting are some of the probabilistic forecasting techniques that have shown accurate results in this space.

VII. CONCLUSION

Long-term load forecasting (LTLF) plays a major role in efficient power system planning and operation. Existing research work experiences limited work done in transmission network long-term tactical planning in power networks that are highly integrated with renewable energy penetration. Nevertheless, this is a key for the purpose of network planning, specially if it is in a transitional stage during construction of new assets. On the other hand, current work have also demonstrated

low accuracy in LTLF results due poor model architectures implemented. Considering these drawbacks, firstly, we have proposed a seq2seq based convolutional neural network and long short term memory (CNN-LSTM) point load forecasting approach that produces accurate forecasting results for a time horizon of three years with a time interval of one month. Identification of most relevant uncertainties such as maximum electricity price, base population change, power consumption, photovoltaic (PV) power generation pattern etc., that have a close relationship with the monthly peak load pattern was a prime step of this work. The outcome of this work has shown its importance for the power network operators, such as: long-term tactical planning up to three years can significantly benefit towards efficient power system operation and one month time interval immensely contribute towards the maintenance scheduling and auxiliary components planning of a power network, such as battery energy storage systems (BESS) and backup generators. As per the best knowledge of authors, this work is the first of its kind in application of CNN-LSTM neural network technique in long-term electricity load forecasting. In addition to point forecasting, probabilistic load forecasting also plays a key role in long-term tactical and strategical planning. Interesting results have been prevailed in this work by implementing a block bootstrapping based long-term probabilistic load forecasting framework.

VIII. FUTURE WORK

It is understood that the probabilistic LTLF framework still has space to be improved by considering different uncertainties that have a strong relationship with the load. In our future work, the probabilistic forecasting model will be improved to overcome the limitations of point forecasting methods and further improve the forecasting accuracy.

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